

Masters Exam: APPLIED PART Answer all 3 questions. Show work in bluebook.

(40 Points) 1) A sports medicine researcher wanted to see if there was a difference in the average pulse rate for college athletes in three sports: basketball, soccer, and tennis. The researcher also wanted to see if such difference might depend on the gender of the athlete. Assume that pulse rates are normally distributed, and that each athlete only plays one sport. The researcher took a random sample of 10 male college students from each sport, and 10 female college students from each sport, and measured the pulse rate for each. The following are the mean and standard deviations for the pulse rates:

	Basketball		Soccer		Tennis	
	Male	Female	Male	Female	Male	Female
Number	10	10	10	10	10	10
Mean	54	60	53	57	67	69
Standard deviation	3.4	3.7	3.9	3.5	3.8	3.6

- Analyze the data to test if there is a difference of pulse rates between the three sports, to test if there is a difference in pulse rates between the two genders, and to test if the difference of pulse rates between the three sports depends on the gender of the athlete.
- Show the model you are using, and estimate the terms in the model. Use the fitted model to estimate the average pulse rate for female tennis players.

(30 Points) 2) A sports researcher wanted to see if there was a difference in the graduation rates for college athletes in three sports: basketball, soccer, and tennis. The researcher also wanted to see if such difference might depend on the gender of the athlete. Assume that each athlete only plays one sport. The researcher took a random sample of 100 male college students from each sport, and 100 female college students from each sport, and observed the number of students who graduated college. The following is the data

	Basketball		Soccer		Tennis	
	Male	Female	Male	Female	Male	Female
Number	100	100	100	100	100	100
Number graduating	69	76	82	83	89	96

Analyze the data to see if there is a difference of graduation rates between the three sports, to see if there is a difference in graduation rates between the two genders, and to see if the difference of graduation rates between the three sports depends on the gender of the athlete.

(30 points) 3) A poultry feed manufacturer has developed a new feed, that the manufacturer claims will increase the mean weight of chicken eggs above the current mean of 58 grams per egg. The manufacturer draws a random sample of 10 eggs and observes the following weights in grams:

58.1 57.9 58.2 58.4 57.8 59.1 59.2 58.9 59.0 58.6

- Assuming that the weight of eggs is normally distributed, is there statistically significant evidence at $\alpha = 0.05$ to support manufacturers claim?
- Assuming that the weight of eggs is normally distributed, give a 95% confidence interval for the average weight of eggs.
- If you cannot assume that the weight of eggs is normally distributed, is there statistically significant evidence at $\alpha = 0.05$, to support the claim that the median weight of chicken eggs under the new feed is greater than 58 grams?